THE CHALLENGE · RECOVER HIDDEN TEXT

The flag had disappeared. Only numbers remained.

SUPPLIED FILE

Dockerfile.json

```
model: "gpt2"
inputs_embeds:
  [ 0.1145, -0.3451, 0.2239, … ]
  [ … ]
  …
```

The first values, rounded for display.

24 groups of numbers

768 numbers per group

Each group represents a piece of text, along with information about its position.

THE TAKEAWAY

Our task: **reconstruct the original text.**

UNDERSTAND · WHAT IS AN EMBEDDING?

A piece of text.
A row in a table.

PIECE OF TEXT

BA

A “token”

→

GPT-2’S PUBLIC TABLE

BA		768 numbers
26		768 numbers
{		768 numbers

Diagram: the bars represent lists of numbers.

This file also adds: THE TOKEN’S POSITION

To distinguish the first, second, third…

THE TAKEAWAY

The text token and its position come from **two known tables.**

A SMALL EXAMPLE · THREE NUMBERS MAKE IT CLEAR

We know what was added.
So we can subtract it.

A token	[2, 5, 1]	+	[1, 0, 3]	Its position
What we receive	[3, 5, 4]	−	[1, 0, 3]	Known position
What we recover	[2, 5, 1]	→	the token	In the dictionary

An invented 3-dimensional example. In the real file, each row contains 768 numbers.

THE TAKEAWAY

Subtraction works one number at a time.

THE BREAKTHROUGH · WE KNOW BOTH TABLES

Subtract the position.
Recover the token.

01

Read a group

the token’s numbers

+ its position’s numbers

The numbers we received.

02

Subtract the position

the token’s numbers

~~its position’s numbers~~

It is in the public table.

03

Compare with the table

BA ✓

The closest token is found.

THE TAKEAWAY

Repeat for all 24 groups, then join the pieces.

VERIFY · RECOVERING TEXT IS NOT ENOUGH

The text was already readable. The distances confirmed it.

DIRECT COMPARISON

3,5 – 9,9

Distance to the dictionary rows.
The correct tokens already emerge.

AFTER SUBTRACTING THE POSITION

0,000 – 0,004

Almost exact matches.
The proposed structure is confirmed.

24 tokens → The same flag in both attempts ✓

THE TAKEAWAY

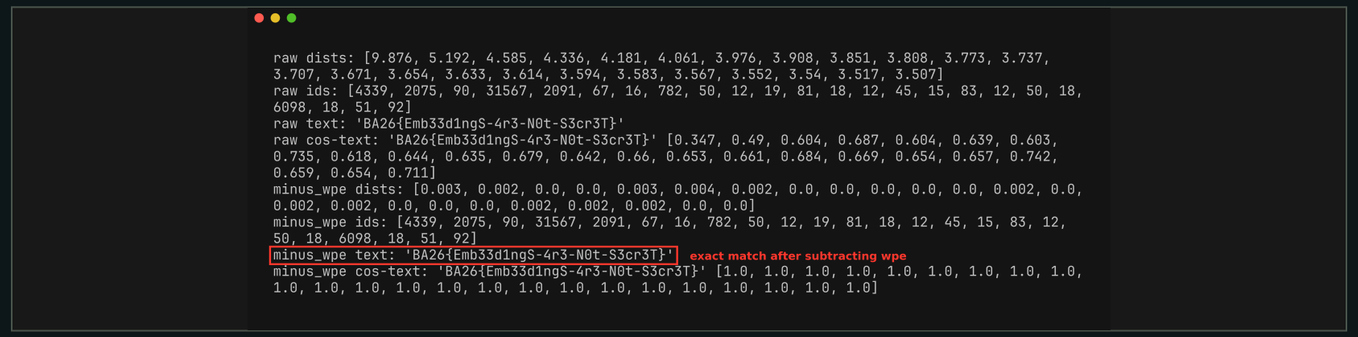
A distance close to zero means: **“we found the same row.”**

THE EVIDENCE · ALL 24 PIECES FIT TOGETHER

The flag put on camouflage. The dictionary recognised it.

BA	26	{	Emb	33	d	1	ng	S	…	}

BA26{Emb33d1ngS-4r3-N0t-S3cr3T}



```
raw dists: [9.876, 5.192, 4.585, 4.336, 4.181, 4.061, 3.976, 3.908, 3.851, 3.808, 3.773, 3.737,
3.707, 3.671, 3.654, 3.633, 3.614, 3.594, 3.583, 3.567, 3.552, 3.54, 3.517, 3.507]
raw ids: [4339, 2075, 90, 31567, 2091, 67, 16, 782, 50, 12, 19, 81, 18, 12, 45, 15, 83, 12, 50, 18,
6098, 18, 51, 92]
raw text: 'BA26{Emb33d1ngS-4r3-N0t-S3cr3T}'
raw cos-text: 'BA26{Emb33d1ngS-4r3-N0t-S3cr3T}' [0.347, 0.49, 0.604, 0.687, 0.604, 0.639, 0.603,
0.735, 0.618, 0.644, 0.635, 0.679, 0.642, 0.66, 0.653, 0.661, 0.684, 0.669, 0.654, 0.657, 0.742,
0.659, 0.654, 0.711]
minus_wpe dists: [0.003, 0.002, 0.0, 0.0, 0.003, 0.004, 0.002, 0.0, 0.0, 0.0, 0.0, 0.0, 0.002, 0.0,
0.002, 0.002, 0.0, 0.0, 0.0, 0.002, 0.002, 0.002, 0.0, 0.0]
minus_wpe ids: [4339, 2075, 90, 31567, 2091, 67, 16, 782, 50, 12, 19, 81, 18, 12, 45, 15, 83, 12,
50, 18, 6098, 18, 51, 92]
minus_wpe text: 'BA26{Emb33d1ngS-4r3-N0t-S3cr3T}'   exact match after subtracting wpe
minus_wpe cos-text: 'BA26{Emb33d1ngS-4r3-N0t-S3cr3T}' [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0,
1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
```

SAVED EVIDENCE · CLICK TO ENLARGE

24 / 24

Tokens recovered.
Almost exact matches
after subtracting the position.

Solved by Claude

THE TAKEAWAY

In this case, turning text into numbers **did not protect its contents.**